

Memory-Efficient Transfer Learning for Airway Segmentation in Ultra-High-Resolution Photon-Counting Computed Tomography

by

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Qualifying Exam Question

Part 1: Automated airway segmentation from computed tomography (CT) is critical for quantitative assessment of pulmonary structure, including applications studying lung disease etiology, early detection, and evaluation of interventions. Most established airway segmentation algorithms, using rule-based, machine learning or deep learning approaches, have been developed and validated using energy-integrating detector CT (EID-CT) chest scans. From the literature, summarize approaches, training strategies, and evaluation methodologies, including quantitative performance metrics, validation datasets, and statistical analysis approaches. Discuss the underlying assumptions of these methods and highlight known limitations and failure modes, particularly those related to image resolution, noise, and generalizability.

Part 2: Photon- counting CT (PCCT) introduces fundamental changes in detector physics, spatial resolution, and noise properties, particularly at ultra-high spatial resolution enabled by 1024x1024 reconstruction matrices. These differences create both opportunities and challenges for translating existing airway segmentation methods to PCCT. Identify the key considerations required to adapt automated airway segmentation algorithms for PCCT. Develop a strategy for PCCT-based airway segmentation, including algorithmic choices, data requirements, and evaluation design. Emphasize how you would assess segmentation performance across reconstruction matrices and airway generations and propose appropriate statistical methods to evaluate performance relative to EID-CT approaches.

Part One

Background: Airway tree segmentation from thoracic computed tomography (CT) plays an important role in quantitative analysis of lung structure. Airway segmentation enables extraction of morphological measurements such as lumen area, inner diameter, and wall thickness. With these, airway segmentation enables assessment of airway disease such as Chronic Obstructive Pulmonary Disease (COPD) [1-3] and bronchiectasis [3, 4].

Accurate, manual airway segmentation is a time-consuming task, with studies reporting segmentation times up to 7 hours [5, 6]. Semi-automatic methods require individual-case interaction and manual correction/validation. Semi-automatic methods range from single seed point placement to user guided local region growing, which can prove similarly time intensive [7].

Fully automatic segmentation alleviates much of the manual effort required for airway segmentation and enables large-scale application. However, the branched morphology of the airway tree is more complex than traditional segmentation tasks such as lung lobes, prostate, and the brain [6, 8]. The intensity profile of the tree is nonuniform, with Hounsfield Unit (HU) values of the airway wall decreasing as airways branch further distally, referred to as “intensity gradient” [5, 9-11]. Distal airway voxels represent much less of the overall segmentation volume than the trachea or bronchi, referred to as “class imbalance”. Intensity gradient, class imbalance, partial volume effects, motion artifacts, and noise may cause the structure to become discontinuous. This challenges many binary segmentation methods which assume a fully connected foreground region [5, 7, 10-14].

To address this, fully automatic airway segmentation literature covers a wide breadth of topics in computer science/engineering and mathematics. These can be grouped into three distinct categories: classical image processing, machine learning, and deep learning-based methods.

Quantification of Results: The most used metric for medical image segmentation similarity is the Dice Similarity Coefficient (DSC), which, like the Jaccard Index (IoU), measures the overlap between two segmentations. Hausdorff Distance is also commonly used, defined as the maximum boundary distance between two segmentations. In airway segmentation literature, this is implemented as the 95th percentile of Hausdorff Distance (HD95) to reduce outlier sensitivity [15]. However, DSC, IoU, and HD95 weigh all voxels equally. Using these metrics, distal airways are weighed significantly less than large airways due to their decreased overall volume [5, 10, 16].

To counteract this, EXACT’09 proposed standard quantifications of segmentation quality compared to a ground-truth reference. Quality is assessed with Tree Length Detected (TLD), Branches Detected (BD), and False Positive Rate (FPR) [7]. Studies have also utilized precision leakage volume and true positive rate (sensitivity) [7, 17-19].

The most common comparison between methods is plotting average TLD against average FPR across a cohort, looking to balance the extent of segmentation against over-segmentation [7]. Segmentation methods are also compared using statistical testing. Authors show statistically significant improvement in TLD, BD, and FPR using paired t-tests for normal distributions [10, 18] and Wilcoxon signed-rank tests for non-normal distributions [5].

Airway segmentation has been benchmarked by several cohorts and challenges. The widely used benchmark for classical image processing methods is EXACT’09 [7]. ATM’22 is most used for modern deep learning approaches [20]. ATM’22 recruited primarily healthy subjects, which may limit the model evaluation under severe airway disease phenotypes. To address this limitation, AeroPath [19] was introduced with the inclusion of COPD and large nodule cases, enabling more comprehensive assessment across disease pathology.

Classical Image Processing: Classical image processing approaches rely on image intensity differences to segment the airways. These methods assume that the airway wall has higher attenuation than the airway lumen and surrounding parenchyma, and that CT resolution is adequate to represent a fully connected airway structure [21].

Early automated airway segmentation by Wood et al. expanded on simple thresholding methods with 3D region growing [22]. Simple thresholding was not viable due to the similar intensity of the airway lumen and surrounding lung parenchyma. Sonka et al. improved region growing methods by incorporating explicit anatomical constraints in 2D to the existing 3D region growing algorithm [23]. Region growing methods were fast but were unable to segment distal airways without leakage. Partial volume effects and low resolution break the underlying assumptions of intensity difference and connectedness in small airways [11, 14].

Fuzzy connectivity was introduced to better model the potentially discontinuous representation of the airway tree. Typically implemented through graph theoretical approaches, fuzzy affinity quantifies pairwise similarity between voxels, and fuzzy connectivity propagates these relationships into a global connectedness. This alters the assumption of binary voxel-wise connection to the tree, allowing partially connected distal airways to be segmented [24]. Leveraging fuzzy connectivity, Tschirren et al. [17] improved TLD and Park et al. [25] reduced FPR relative to previous region growing methods. However, fuzzy connected approaches still used the assumption that the intensity of the airway wall is significantly higher than that of the lumen and parenchyma. Fuzzy connectivity methods remained limited by the intensity gradient and noise/partial volume effects of distal airways.

Morphological and geometric approaches aimed to assess the shape of the tree structure along with intensity differences. Kiraly et al. addressed leakage in distal airways by leveraging geometric constraints around the lumen [14]. Pu et al. addressed leakage with curvature analysis, removing classified spherical and planar regions which were likely over-segmentation [26]. However, these methods rely on assumptions of airway morphology and image quality. As distal airways approach the spatial resolution of CT or deviate from the expected tubular anatomy, the reliability of these methods decreases and leads to leakage or missed branches [14].

Across region growing, fuzzy connectivity, and morphological methods, a shared underlying assumption is that the airway lumen can be reliably separated from the surrounding tissue using local intensities. This creates the main failure point of classical image processing, distal airways. Small airways demonstrate reduced lumen/parenchymal contrast. This is compounded by noise and partial volume effects, breaking the underlying assumption of the methods.

Machine Learning: Machine learning enables segmentation of distal airways by shifting from intensity-based assumptions to supervised, manually crafted features. The core assumption of machine learning is that the airway and non-airway voxels may be separated with a finite set of predefined features. In practice, these features include curvature [27-29] and radiomic texture features [27] which encode tubular structure and intensity patterns, respectively.

Machine learning methods demonstrated superior ability to segment distal airways from classical image processing. Meng et al. used curvature features for a support vector machine, improving FPR [28]. Bian et al. used curvature and radiomic features in a random forest classifier, improving BD [27]. This supported the assumption that hand-crafted features may capture the difference between airway and non-airway voxels within a training/validation set.

However, neither approach was validated on an external dataset. This highlights the key failure mode of machine learning based airway segmentation, its reliance on the stability of the features selected. Radiomic features are known to vary with CT acquisition/reconstruction protocol and require harmonization [30, 31]. Without harmonization, when applied to a changing protocol, the assumption that the airway and non-airway voxels may be separated using solely the selected features may no longer hold.

There is limited literature reviewing this domain shift in airway segmentation. This is likely due to the parallel development of deep learning approaches, which quickly became dominant in medical image segmentation.

Deep Learning: Early deep learning methods have been shown to outperform classical image processing methods in TLD, BD, and FPR [18, 19, 32]. Deep learning relies on the assumption that the airway and non-airway voxels may be separated by a hierarchy of feature representations derived from the data, and that these representations may generalize to unseen data.

The dominant architecture in medical image segmentation is the U-Net, an encoder-decoder neural network which uses skip-connections to preserve high resolution detail while incorporating increasingly abstracted features [33]. This architecture utilizes a convolutional neural network (CNN), where learned convolution kernels extract features from the image [34]. This has been successfully implemented in airway segmentation, leveraging the U-Net's multi-scale architecture to segment distal airways [6, 18].

Importantly, deep learning methods can incorporate domain knowledge from classical image processing into existing architectures. The U-Net has been combined with region growing [12], graph theory [35], and airway morphology [6, 36]. Further developments have created bronchiole-specific loss functions [9] and enhanced multi-scale analysis [16].

Other methods have been implemented for deep learning-based airway segmentation such as fully convolutional [11] and generative-adversarial methods [13, 37]. However, the U-Net backbone is most widely used and successful architecture.

Deep learning models are highly impacted by hyperparameter selection. The most common loss functions used for airway segmentation are Binary Cross-Entropy Loss (BCE) and DICE Loss [18, 35]. Due to the limitations of DICE score previously mentioned, BCE is more commonly used. Several groups have created loss functions tailored to airway segmentation. Notably, Zheng et al. proposed General Union Loss to address class imbalance by implementing distance-based weighting [10]. Approaches overwhelmingly minimized loss with the ADAM optimizer [38] and initial learning rate of 1×10^{-4} [5, 6, 18, 32, 33, 39, 40]. ReLU activation functions are commonly used through standard ReLU [32] or leaky ReLU variations [5].

Deep learning is susceptible to overtraining. This is alleviated in airway segmentation with data augmentation techniques including elastic deformation [9, 33, 39], random cropping [18], random patches [18], random rigid transformation [5, 9, 10, 32], random noise [9], and random blur [9].

With data augmentation methods and sufficient training data, deep learning methods for airway segmentation have demonstrated adequate generalizability across variations in acquisition protocol [40] and external datasets [19]. This performance supports the underlying assumption that the hierarchical feature representations found through CNNs can separate airway and non-airway voxels and can generalize to unseen data. The U-Net specifically has been able to circumvent the assumptions that plateaued classical image processing and machine learning methods. The leading benchmarks of the ATM'22 challenge are variations of the U-Net architecture utilizing classical image processing knowledge rather than a separate architecture backbone [20].

Part Two

Photon Counting CT: Photon Counting Computed Tomography (PCCT) is a recent development in CT technology enabling increased resolution, elimination of electronic background noise, and equivalent image quality with reduced dose [41, 42]. The advancements from PCCT contribute greatly to quantitative (q)CT, specifically by the ultra-high-resolution (UHR) acquisition mode.

UHR acquisition allows image matrices to be reconstructed to 1024^2 pixels in-plane (1024^2 PCCT) or 512^2 pixels in-plane (512^2 PCCT), opposed to only 512^2 pixels in-plane in the standard energy-integrating CT reconstruction (512^2 EID-CT). UHR PCCT also enables thinner slice thickness, reduced from 0.5-0.6mm in EID-CT [43] to 0.2mm in PCCT [43, 44].

The core change in PCCT is the photon counting detector. Instead of the typical two-stage scintillation and photodiode-conversion, PCCT utilizes a Cadmium Telluride or Cadmium Zinc Telluride crystal to directly convert photons to electron-hole pairs. A high voltage (800V-1kV) is applied to the detector crystal, with a common cathode and pixelated anode. Converted electron-hole pairs travel to the nearest anode upon impact, allowing much smaller detector pixels than achievable in EID-CT [43, 45].

UHR subsamples the 4×4 detector pattern used by the Macro and Chess PCCT acquisition modes, instead grouping 2×2 detector pixels into one sinogram intensity [45]. While this allows finer sampling of the anatomy, typical CT physical limitations such as charge sharing and PCCT-specific effects such as pulse pileup are amplified [45]. The physical changes of the system, along with the increased resolution, have been shown to impact pulmonary structural visualization [2, 46, 47], noise profile [42], modulation transfer function (MTF) [41], and qCT metrics of the airways [2, 44, 48].

The changes in detector physics and image resolution have shown promise to increase the extent of airway segmentation in PCCT, notably through more accurate wall thickness measurement and smaller airway detection [48, 49]. However, no airway segmentation method has been created explicitly for 1024^2 PCCT. Development of a method optimized for 1024^2 PCCT is critical to fully utilize the capability of UHR acquisitions in qCT of the lungs.

Limitations to Existing Methods: The largest limitation to airway segmentation in 1024^2 PCCT is the size of the images and resulting computational complexity of analysis. To demonstrate image size, our group found that the mean inferior-superior extent of the thoracic cavity is 301mm in 904 subjects acquired from COPDGene [1, 50]. To fully sample this region with 512^2 EID-CT at 0.5mm slice thickness, 158 million voxels are required. For the equivalent coverage using 1024^2 PCCT at 0.2mm slice thickness, 1.58 billion voxels are required. The change in image size represents a significant increase in computational complexity for airway segmentation systems which scale nonlinearly.

Existing deep learning methods commonly leverage 3D convolutions for inter-slice anatomical context. However, the ten-fold increase in image size with 1024² PCCT creates a computational challenge for these architectures. Assuming isotropic voxels, the computational complexity of a 3D convolution scales proportionally with its input image size (N_x, N_y, N_z), kernel volume (k^3), and channels (C_{in}, C_{out}) [34]. Quantifying computational complexity in floating-point operations (FLOPs) performed by the graphics processing unit (GPU):

$$3D \text{ Convolution FLOPs} \propto N_x N_y N_z k^3 C_{in} C_{out}$$

In contrast, 2D convolutions only scale with the in-plane image dimensions.

$$2D \text{ Convolution FLOPs} \propto N_x N_y k^2 C_{in} C_{out}$$

In a review on convolution methods in 512² EID-CT, Zhang et al. found an average of 1.81x increase in peak GPU memory required, 4.14x increase in training time, and a 2.43x increase in inference time from 2D to 3D convolutions [51]. In several studies, 2D CNNs outperformed 3D CNNs due to accommodations in the 3D architecture for GPU memory limitation [52-55]. Non-patch-based 3D convolutions would not be able to account for the memory requirements when assessing 1024² PCCT.

While the 2D U-Net provides decreased memory intensity, their lack of inter-slice awareness limits effectiveness in airway segmentation. Context in the anterior-posterior direction is essential for small airway detection due to noise and partial volume effects. This was directly demonstrated by Yang et al., who compared 2D and 3D U-Nets in airway segmentation [16].

Current state-of-the-art airway segmentation methods rely on patch-based training to limit 3D convolution GPU memory [18]. Despite patching, these methods remain computationally intensive [51]. In 1024² PCCT, a 3D patch covering the same anatomical area from 512² EID-CT would require approximately ten times more voxels. Consequently, either minimum GPU requirements would rise substantially, or anatomical coverage of each patch would be greatly reduced.

2.5-Dimensional Convolution: The underperformance of 2D convolutions and the high memory use of 3D convolutions creates the need for a hybrid approach between the two for use in 1024² PCCT airway segmentation. Zhang et al. and Li et al. conducted reviews on 2D, 3D, and hybrid 2.5D approaches [51, 53]. 2.5D convolution is an umbrella term for three separate uses of 2D convolution leveraging 3D inter-slice information containing: 1) 2.5D Multi-Slice, 2) 2.5D Multi-View, and 3) 2.5D Hybrid 2D-3D [56]. 2.5D Hybrid 2D-3D approaches demonstrate similar peak GPU usage to pure 3D approaches and will not be pursued for this study [51].

2.5D Multi-Slice aims to analyze the image using parallel, adjacent planes as context for the slice being analyzed. The middle plane is used as the primary convolution plane, with neighboring planes as additional channels. This scales the computation linearly by the number of context slices used (N_{slice}).

$$2.5D \text{ Multi-Slice Convolution FLOPs} \propto N_x N_y k^2 N_{slice} C_{in} C_{out}$$

2.5D Multi-View aims to assess three intersecting orthogonal planes separately, then combine the segmentations. The method uses one standard 2D U-Net for each plane, followed by a separate network to combine them. Zhang et al. recommend a lightweight 3D Volumetric Fusion Network (VFN) for orthogonal image fusion [15]. Each plane is analyzed in series then the VFN combines them, enabling peak GPU usage comparable to or lower than pure 2D approaches [51].

$$2.5D \text{ Multi-View Convolution FLOPs} \propto (3 \times 2D \text{ Convolution FLOPs}) + VFN \text{ FLOPs}$$

Only one publication has targeted automated airway segmentation with 2.5D convolutions. In 2019, Yun et al. utilized the 2.5D Multi-Slice architecture to train a U-Net in 512² EID-CT [39]. The method was compared to the current state-of-the-art [11] with DSC, TLD, BD, and FPR. The architecture demonstrated lower inference time at the expense of segmentation quality, GPU usage was not discussed.

2.5D Multi-Slice and 2.5D Multi-View both offer reduced memory complexity and may be assessed for 1024² PCCT airway segmentation. However, 2.5D Multi-Slice loses the spatial order of the adjacent slices, instead using them as independent channels. This may decrease segmentation quality, as seen in Yun et al. [39]. 2.5D Multi-View will be pursued due to the significance of inter-slice relationships in airway segmentation and its decreased peak GPU use from 2.5D Multi-Slice.

Transfer Learning: Existing airway segmentation cohorts do not include labeled PCCT data. Because of this, transfer learning is warranted. Atasever et al. define a transfer learning decision framework for medical image segmentation [57]. From this, a novel airway segmentation method can be developed with a homogeneous, inductive, parameter-based transfer learning approach. Initial weights and parameters can be developed from large 512^2 EID-CT datasets, and parameter fine-tuning can be performed on less-available 1024^2 PCCT data.

Shallow layers of a network capture lower-level features, while deeper layers detect increasingly abstract and task-based information [57]. Although EID-CT and PCCT demonstrate differences in image characteristics, I assume that the underlying airway morphology remains consistent between the systems. Under this assumption, the shallow layers of the network are expected to retain large-scale anatomical feature representations between domains, while deeper layers may require adaptation to PCCT-specific image characteristics. Freezing shallow, transferrable layers reduces the number of trainable parameters, potentially reducing the amount of target-domain data required [57].

Aim 1: I hypothesize that a 2.5D Multi-View U-Net architecture, initially weighted on 512^2 EID-CT and fine-tuned on 1024^2 PCCT, will outperform a selected state-of-the-art method in TLD, BD, and FPR when applied to 1024^2 PCCT data. Aim 1 will create a method optimized for 1024^2 PCCT airway segmentation and provide a baseline for future techniques.

Aim 2: I hypothesize that 512^2 PCCT ground-truth segmentations will show improved TLD, BD, and FPR compared to matching 512^2 EID-CT ground-truth segmentations. TLD, BD, and FPR for both methods will be relative to a 1024^2 PCCT ground-truth segmentation. Aim 2 will quantify the effect of changing detector technology on airway segmentation, independent of resolution.

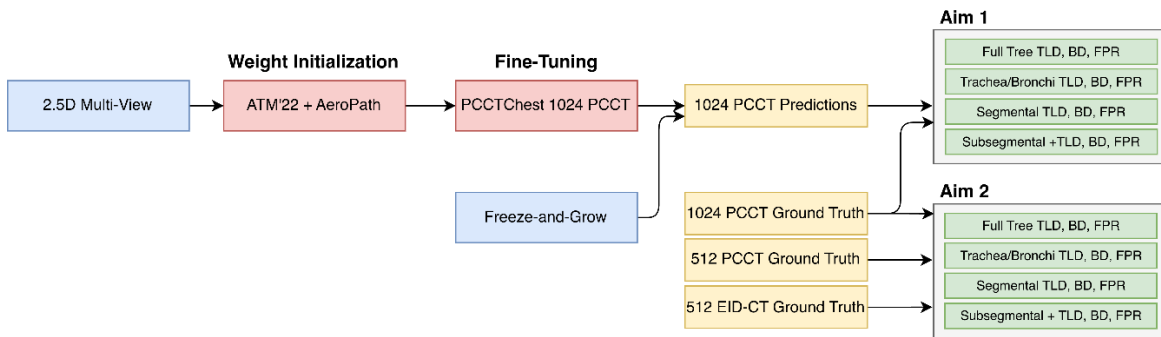


Figure 1. Evaluation pipeline. Blue boxes represent airway segmentation methods. Red boxes represent training on the labeled dataset using the pipeline shown in **Figure 2**. Yellow boxes represent images from the labeled dataset. Green boxes represent statistical comparisons.

Implementation: The wave front propagation used for assessing BD in EXACT'09 inherently divides and categorizes airway branches into their respective generations [7]. I will assess TLD, BD and FPR of the full tree, generation 0-2 (trachea-bronchi), generation 3 (segmental bronchi), and generation 4+ (subsegmental bronchi and further) separately. This comparison will assess performance of the system as airway diameter decreases relative to spatial resolution. Sub-subsegmental bronchi may be included as a unique category after initial results are gathered.

While not optimized for PCCT, Freeze-and-Grow's hybrid region growing and deep-learning correction architecture has demonstrated ability to segment 1024^2 PCCT images in our lab [12]. Freeze-and-Grow will be used as the state-of-the-art comparison method for Aim 1.

A dataset (PCCTChest) of healthy volunteers and COPDGene subjects is currently being recruited by our group [1]. The subjects are acquired with proposed optimal acquisition and reconstruction parameters for qCT of the lungs [44, 58]. Each subject undergoes 512^2 EID-CT, 512^2 PCCT, and 1024^2 PCCT.

I will train the network using General Union Loss for distance-based weighting of segmentation similarity [10]. Loss will be minimized with the ADAM optimizer and an initial learning rate of 1×10^{-4} . Overtraining will be reduced with data augmentation via random cropping, random elastic deformation, and random rigid transformation. I will not apply random noise or blur, as an aim of this study is to assess segmentation quality variation due to noise profile changes from EID-CT to PCCT. Due to limited sample size in PCCTChest (current $N=10$), leave-one-out cross-validation will be used. As more subjects are recruited, dedicated training/validation/testing splits will be used.

Segmentation metrics will be analyzed with pairwise statistical methods through GraphPad Prism. I will use a Shapiro-Wilk test for normality. This will be followed by a paired t-test in normally distributed groups and a Wilcoxon signed-rank test in non-normally distributed groups, as previously used in the literature [5, 10, 18]. Significance will be achieved with a P-value less than 0.05.

PCCTChest ground-truth labels have not been developed. However, semi-automated methods such as VIDA Insights for 512² EID-CT images have been extensively used by our group [59]. I propose initial segmentations of 512² PCCT and 512² EID-CT using VIDA Insights. An affine transform will then be calculated by registering the subject's 512² PCCT image to the subject's 1024² PCCT image. The final transform, optimized with mean squared error, will be applied to the segmentation image using nearest-neighbor interpolation. This will transform the segmentation to the corresponding 1024² PCCT physical space. Transformed segmentations will be manually corrected and extended to higher generation airways than detectable with 512² PCCT.

To address inter- and intra-observer variability, I will repeat a manual segmentation sub-study used by our group in lung nodule segmentation [31, 60]. This study required DSC over 0.80 for identical cases between researchers or repeated by the same researcher. Acknowledging the weakness of DSC for airway segmentation, I propose a similar sub-study using TLD, BD, and FPR, initially targeting 90% similarity (TLD, BD) and 5% error (FPR), which may be adjusted after visual inspection.

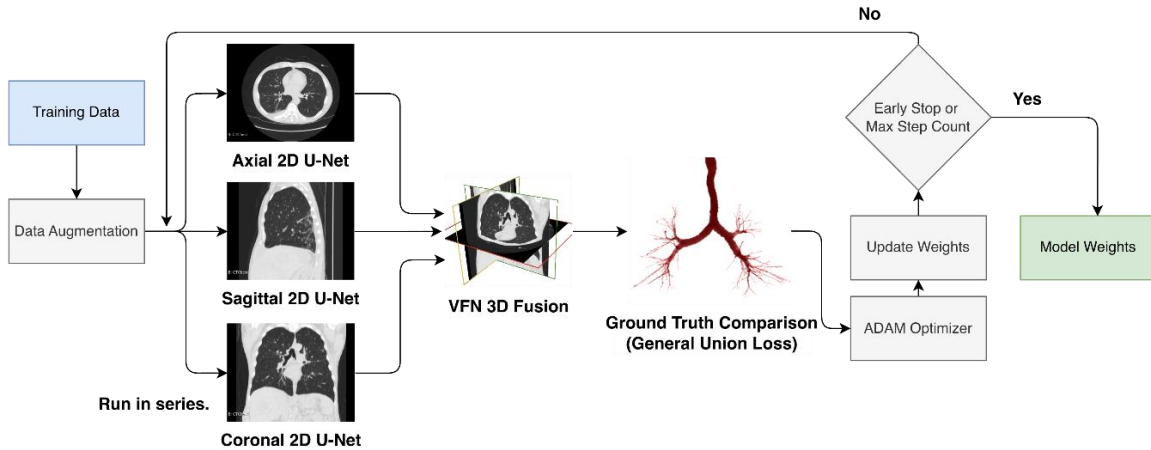


Figure 2. Training pipeline for 2.5D Multi-View U-Net architecture.

Limitations and Future Work: While actively recruiting, the current PCCTChest dataset is small. We have recruited 8 COPD subjects and 2 healthy volunteers. Further, the number of healthy volunteers compared to COPD subjects may bias the network toward diseased lung morphology. Significant dataset expansion or further external PCCT/EID-CT paired datasets are essential for statistical validity of our results.

Patch-based approaches may further reduce the computational cost of the network. This will not be implemented due to the already confounding nature of the inter-scanner differences and changing resolution. Patch-based processing will be implemented in the future.

Our transfer learning approach assumes that domain differences between PCCT and EID-CT are weighed only in the deepest layers of the network. With additional training data, shallow layers may be unfrozen to investigate the network's representation of anatomical features. Further, it is assumed that the parameters alone may capture the change from EID-CT to PCCT. Feature-based transfer learning may be used to explore how the network represents the change in detector physics and image resolution.

While the acquisition and reconstruction protocol used in PCCTChest has been recommended for lung texture analysis, Sieren et al. suggested sharper kernels may be used for airway-specific analysis [44]. Future work may be conducted utilizing ReconCT to reconstruct stored raw data with a sharper kernel for repeated PCCT parameter fine-tuning.

Finally, a standard U-Net architecture was chosen specifically for its simplicity compared to contemporary airway segmentation methods. This simplicity will allow for accurate comparison of future systems with a 1024² PCCT-trained baseline.

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